

PRACTICAL 4

Simulate the following AR (2) series:

- 1) $y_t = 0.4y_{t-1} + 0.5y_{t-2} + \varepsilon_t$
- 2) $y_t = 1.2y_{t-1} - 0.7y_{t-2} + \varepsilon_t$

Plot the correlogram, PACF, time series plot and check for stationarity.

Code & Output

```
#To simulate AR(2) series  
y1=arima.sim(n=1000,model=list(order=c(2,0,0),ar=c(0.4,0.5)))
```

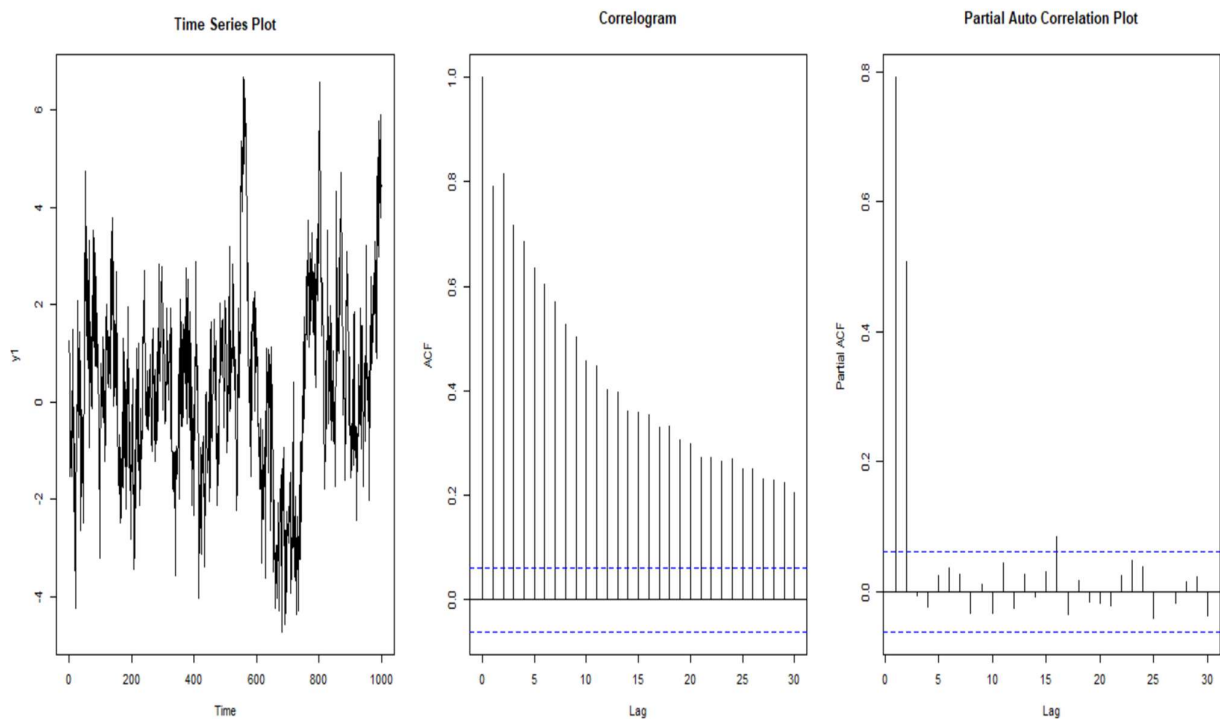
```
par(mfrow=c(1,3))  
plot.ts(y1,main="Time Series Plot")  
acf(y1,main="Correlogram")  
pacf(y1,main="Partial Auto Correlation Plot")  
adf.test(y1)
```

Augmented Dickey-Fuller Test

data: y1

Dickey-Fuller = -4.6932, Lag order = 9, p-value = 0.01

alternative hypothesis: stationary



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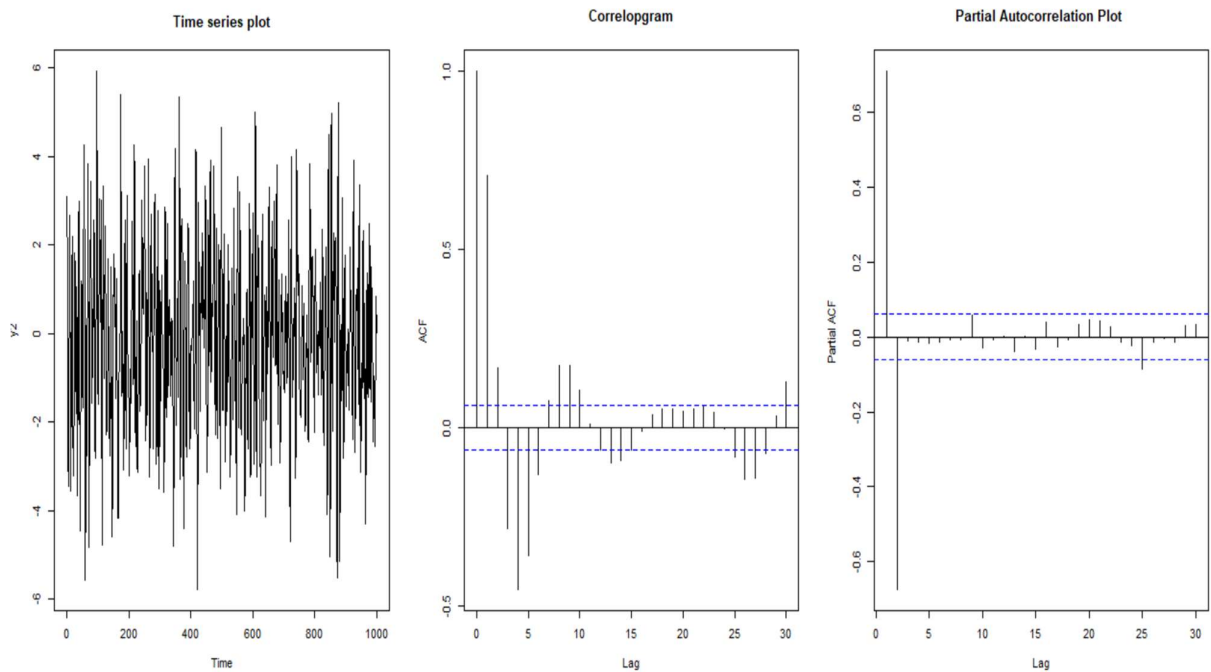
```
y2=arima.sim(n=1000,model=list(order=c(2,0,0),ar=c(1.2,-0.7)))  
ts.plot(y2,main="Time series plot")  
acf(y2,main="Correlogram")  
pacf(y2,main="Partial Autocorrelation Plot ")  
adf.test(y2)
```

Augmented Dickey-Fuller Test

data: y2

Dickey-Fuller = -10.28, Lag order = 9, p-value = 0.01

alternative hypothesis: stationary



PRACTICAL 5

Simulate the following MA series:

1) $y_t = \varepsilon_t + \varepsilon_{t-1}$

2) $y_t = \varepsilon_t + 0.4 \varepsilon_{t-1} + 0.22 \varepsilon_{t-2}$

Plot the correlogram, PACF, time series plot and check for stationarity.

Code & Output

```
y1=arima.sim(n=1000,model=list(order=c(0,0,1),ma=1))
```

```
par(mfrow=c(1,3))
```

```
plot.ts(y1,main="Time Series Plot")
```

```
acf(y1,main="Correlogram")
```

```
pacf(y1,main="Partial Auto Correlation Plot")
```

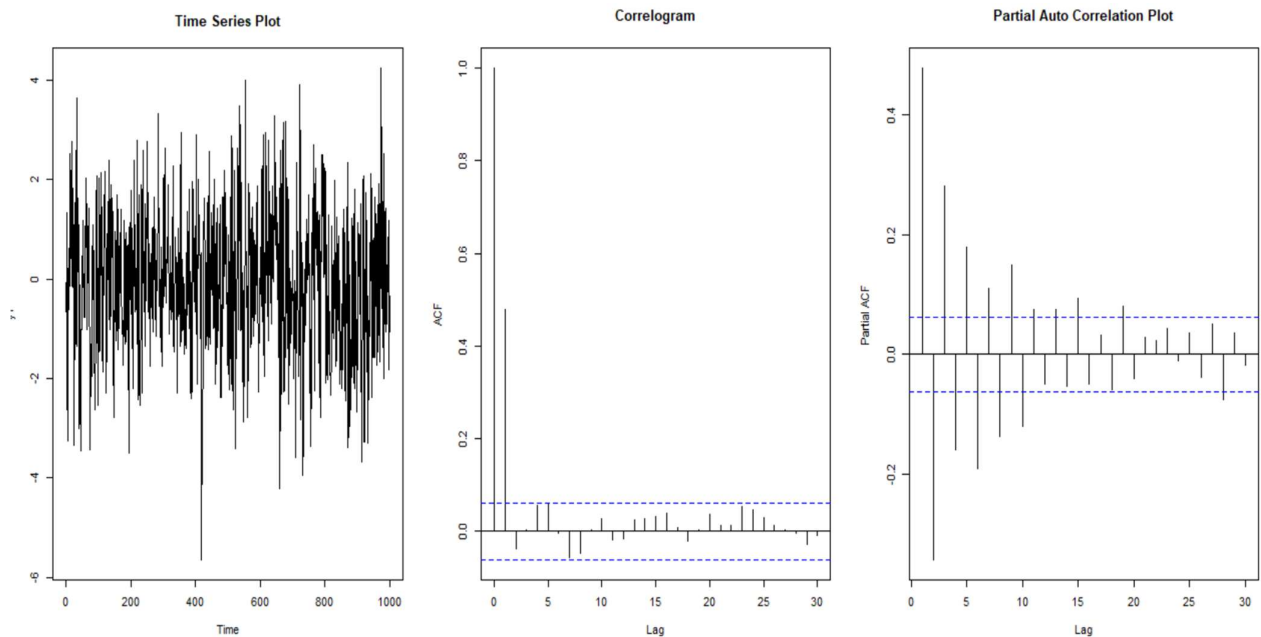
```
adf.test(y1)
```

Augmented Dickey-Fuller Test

data: y2

Dickey-Fuller = -9.7744, Lag order = 9, p-value = 0.01

alternative hypothesis: stationary



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```
y2=arima.sim(n=1000,model=list(order=c(0,0,2),ma=c(0.4,-0.22)))  
ts.plot(y2,main="Time series plot")  
acf(y2,main="Correlogram")  
pacf(y2,main="Partial Autocorrelation Plot ")  
adf.test(y2)
```

Augmented Dickey-Fuller Test

data: y2

Dickey-Fuller = -11.108, Lag order = 9, p-value = 0.01

alternative hypothesis: stationary

